

Different Generations in Cellular Communication System

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Introduction to 5G & 6G :-

- ⇒ A Cellular Network or mobile Network is a wireless communication system that divides a geographic area into sections, called Cells.
- ⇒ Each cell is served by a base station.
- ⇒ Cell networks allow mobile devices to connect to the internet, make phone calls & send/receive texts etc.

Different Generations :-

- (1) 1-G :-
- ① Mobile Communications was first introduced in 1980s & Completed in early 1990s.
 - ② speed was upto 2.4 Kbps
 - ③ operating frequency was 800-900 MHz.
 - ④ It used Analog Signals
 - ⑤ Coverage area was small.
 - ⑥ only voice can be transmitted & its quality was low
 - ⑦ No roaming support between various operators.

- (2) 2-G:-
- ① 2-G was shifted from Analog to Digital.
 - ② GSM (Global system for Mobile Communication) technology was introduced.
 - ③ Voice call was upgraded to mms (multimedia messaging system) & sms (short message system).
 - ④ operating freq = 1.8 GHz
 - ⑤ Data speed = 64 Kbps
 - ⑥ It supported both voice & data transmission but couldn't handle video transmission.
 - ⑦ 2G + GPRS (General Packet Radio Service) = 2.5G was introduced thereafter.

- (3) 3-G:-
- ① Data transfer technology used was UMTS (Universal Mobile Terrestrials system).
 - ② It supported both voice & data
 - ③ Data speed increased to 3.1 Mbps
 - ④ We had option of Video calling + play games on mobile phones.
 - ⑤ It offered highspeed wireless internet along with international roaming.

- ④ 4-G :- ① In 4-G, LTE (Long Term Evolution) & Wi-fi (wireless fidelity) data technology was used. ③
- ② Vo-LTE (Voice over LTE) is for both voice & internet.
- ③ 4-G supports HD quality streaming
- ④ speed = 100 Mbps
- ⑤ The internet service provided by Ultra-Broadband
- ⑥ It supports Video Conferencing.

- ⑤ 5-G :- ① 5-G supports secure & fast Connectivity.
- ② 5-G integrates with A.I
- ③ It is faster than 4-G.
- ④ It has ability to have multiple people in Video Conferencing calls.
- ⑤ Seamless Connectivity across all locations like ocean, remote, air etc.
- ⑥ It has massive network Capacity.
- ⑦ Speed of real time applications is fast
eg:- Maps & Weather.

6-9:- ① It has extremely high data rates;
Speed = 1 Tbps (Tera bit per second)

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② It enables real time holographic communication & ultra HD streaming

③ Ultra Low latency (0.1 msec)

④ Use of (THz) Terahertz frequencies

⑤ It supports for advanced applications like extended reality, Metaverse, Brain-Computer interfaces etc

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